

Big Data: debunking some of the myths

Chris Swan
@cpswan

Agenda

- My background
- What do I mean by big data?
- Know your algorithm
- Know your data
- Performance

My background



CTO



CTO Client Experience
Co-head CTO Security



Corporate Finance
fintech, early stage

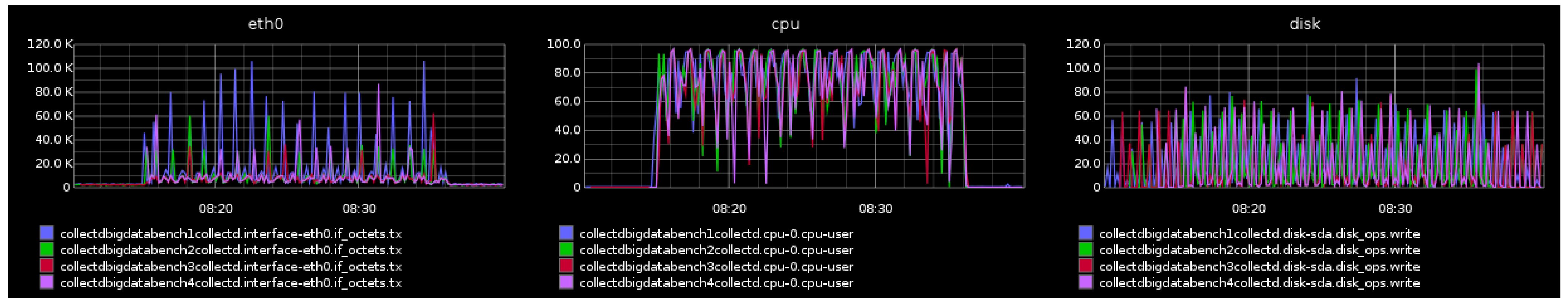


IT R&D – Networks and security
Grid, app server engineering



Combat System Engineer

Recent adventure with Big Data



Misquoting Roger Needham

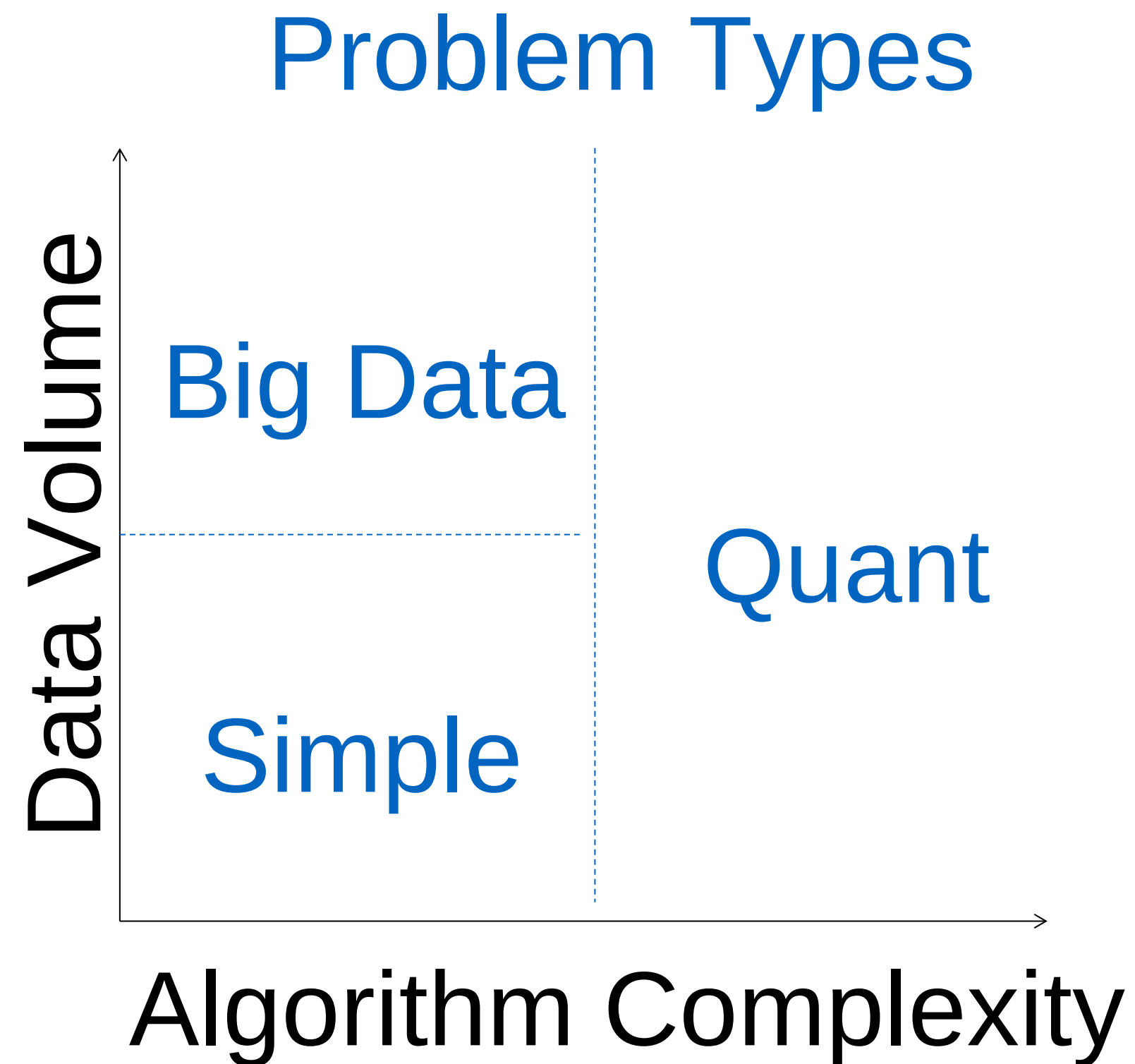
*Whoever thinks their analytics
problem is solved by big data,
doesn't understand their analytics
problem and doesn't understand
big data*



What do I mean by 'big data'?

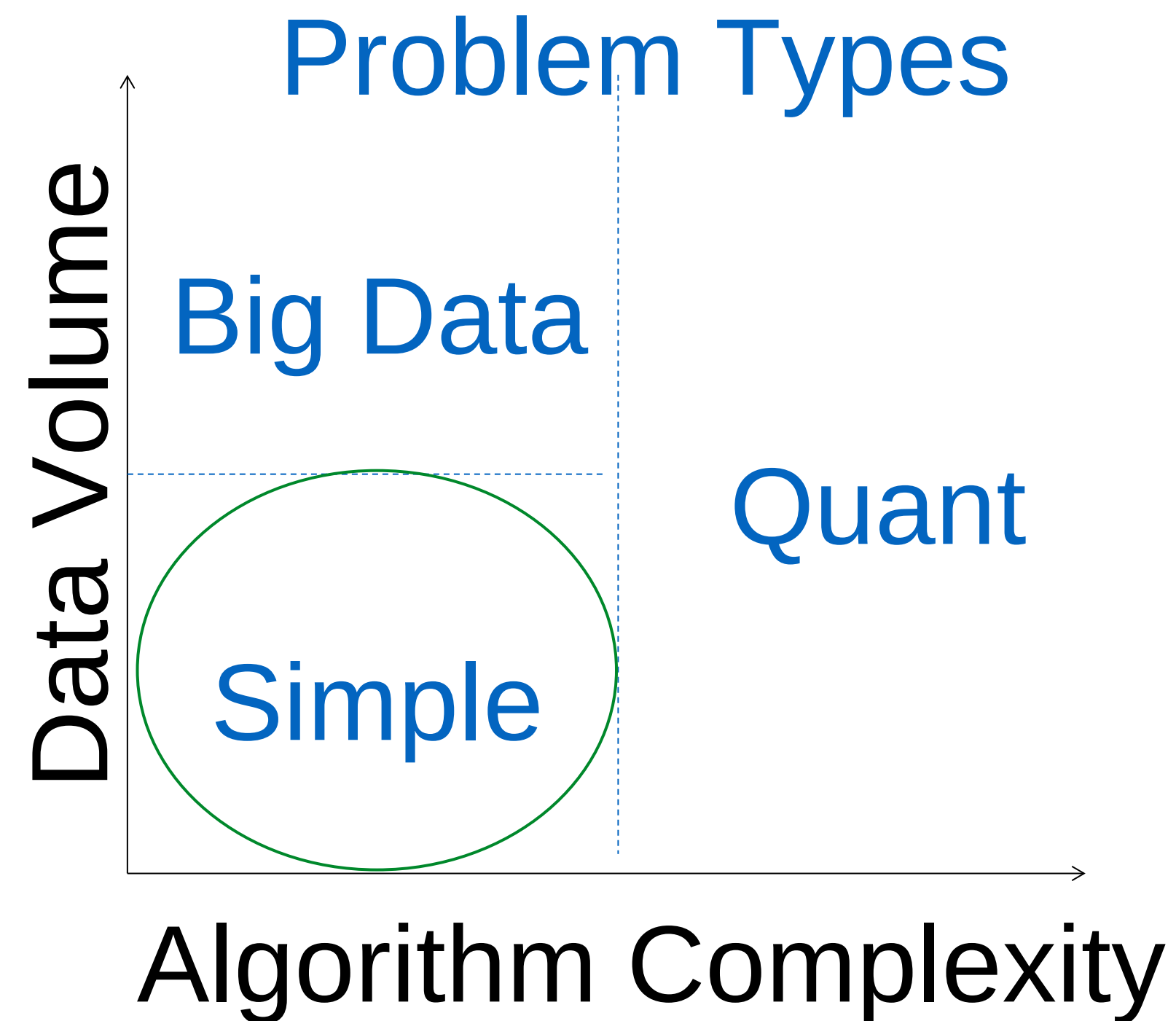
Overview

Based on a blog post from April 2012 – <http://is.gd/swbdla>



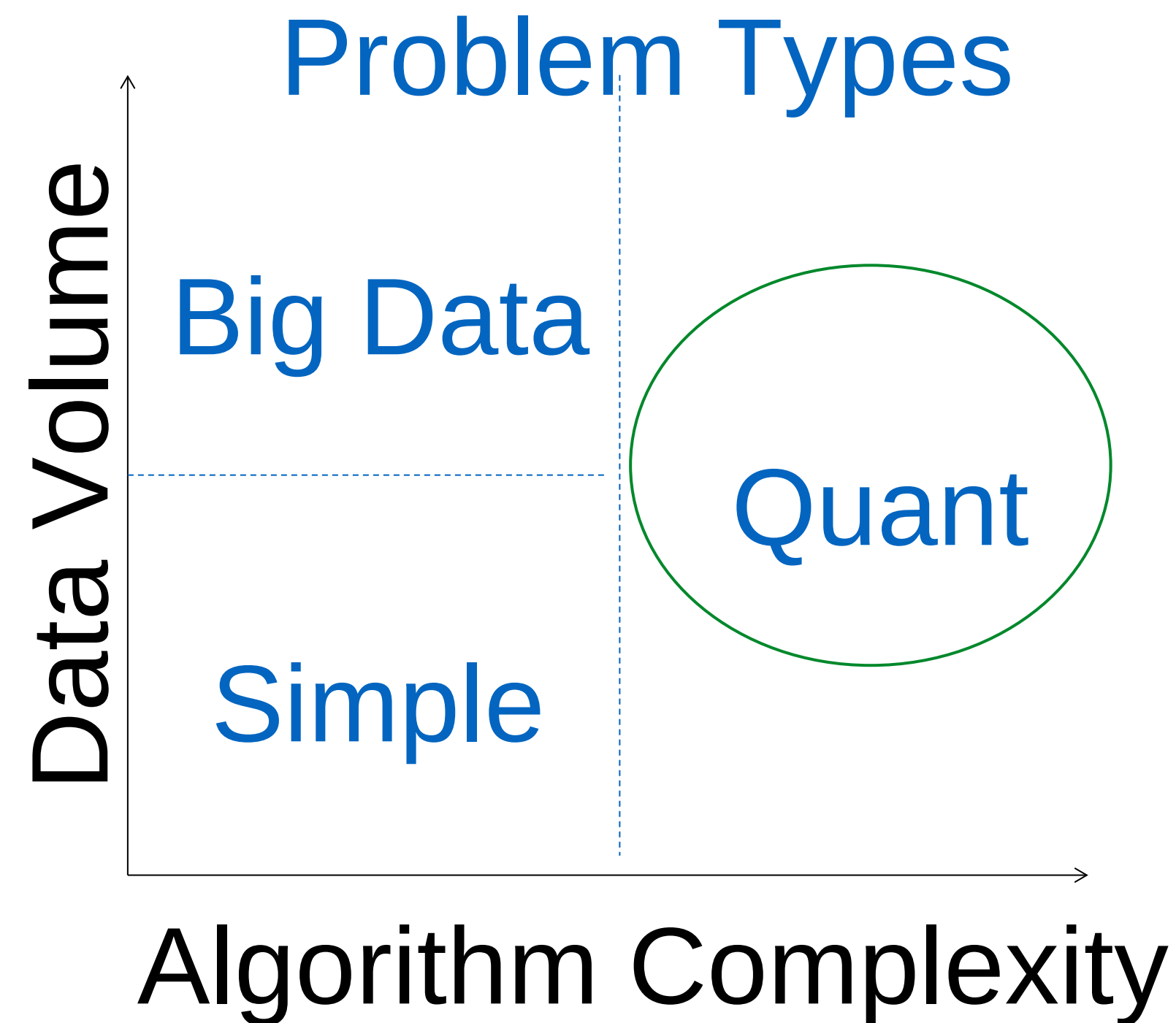
Simple problems

Low data volume, low algorithm complexity



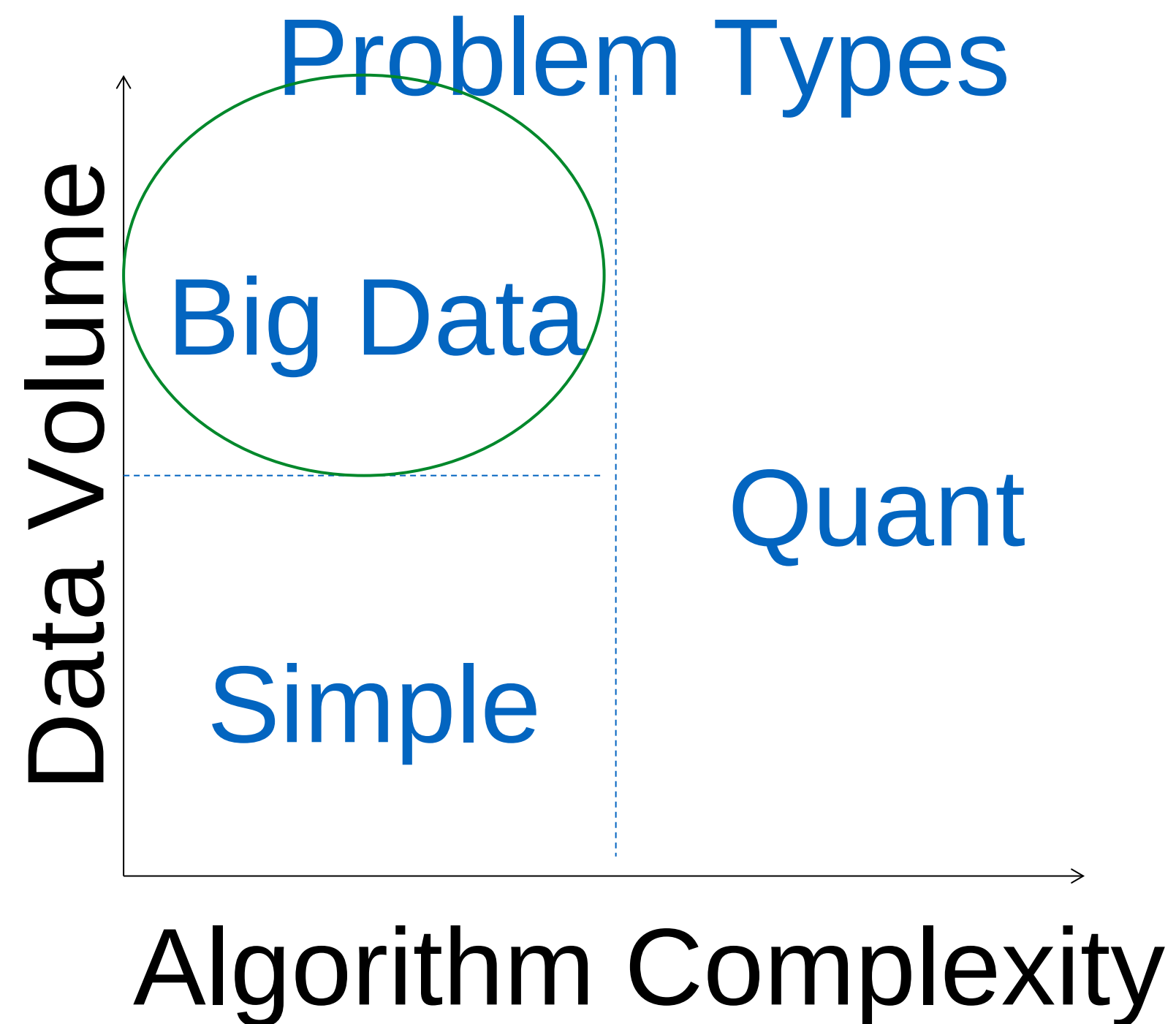
Quant Problems

Any data volume, high algorithm complexity



Big Data Problems

High data volume, low algorithm complexity



Types of Big Data Problem:

1. Inherent
2. More data gives better result than more complex algorithm

The good, the bad and the ugly of Big Data

Good

- Lots of new tools, mostly open source



Bad

- Term being abused by marketing departments

Ugly

- Can easily lead to over reliance on systems that lack transparency and ignore specific data points
'Computer says no', but nobody can explain why

It's important to know your algorithms

Turning an assumption into a line

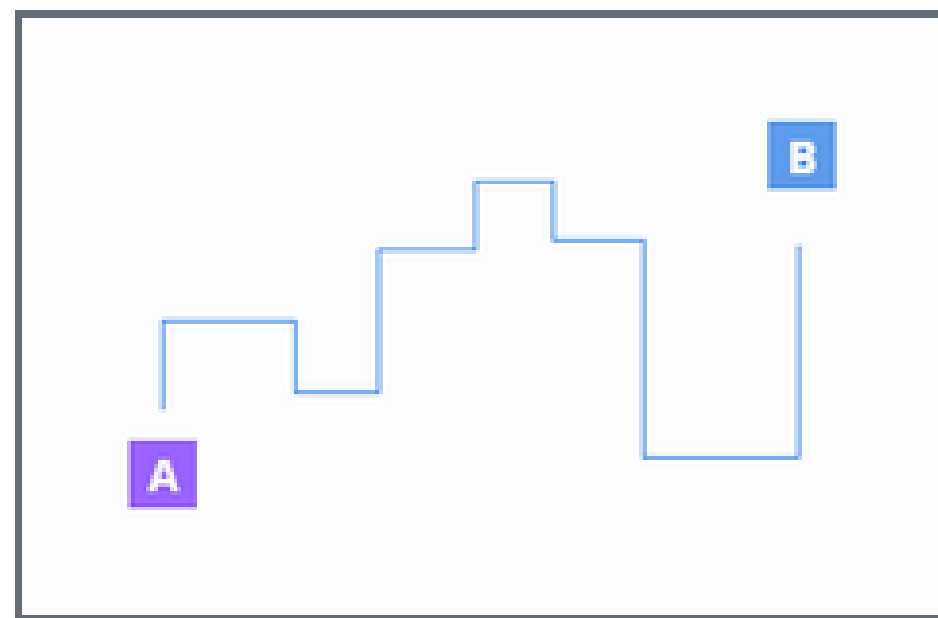


There are lots of algorithms to understand

Algorithmia Launches With More Than 800 Algorithms On Its Marketplace

Posted Mar 12, 2015 by [Kyle Russell \(@kylebrussell\)](#)

Featured Demos



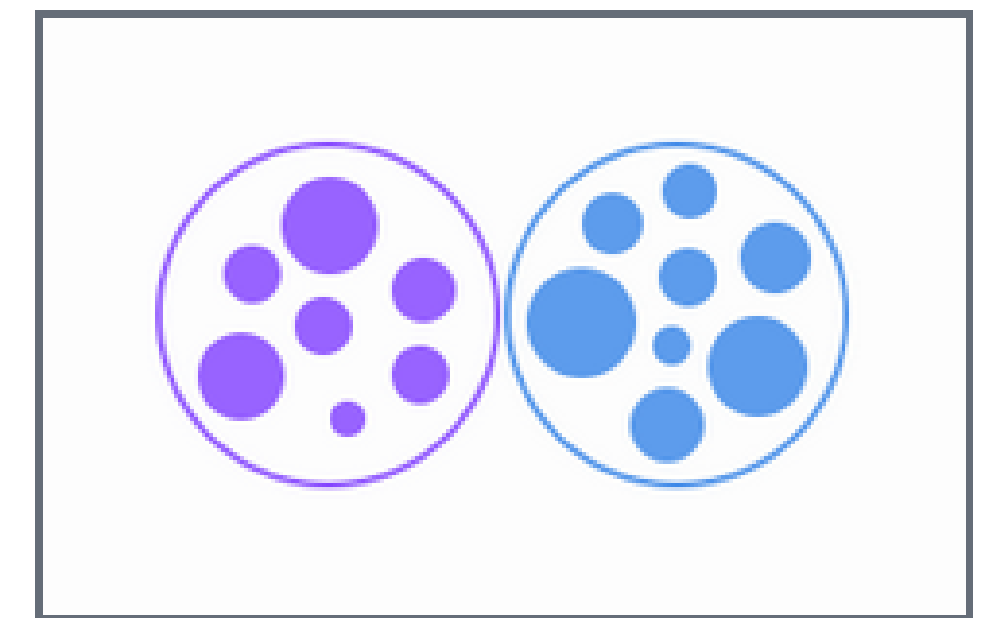
Path Planning

Plan with [Dijkstra's algorithm](#).



Auto Tag Urls

Generate [tags](#) for any URL.



Topic Analysis

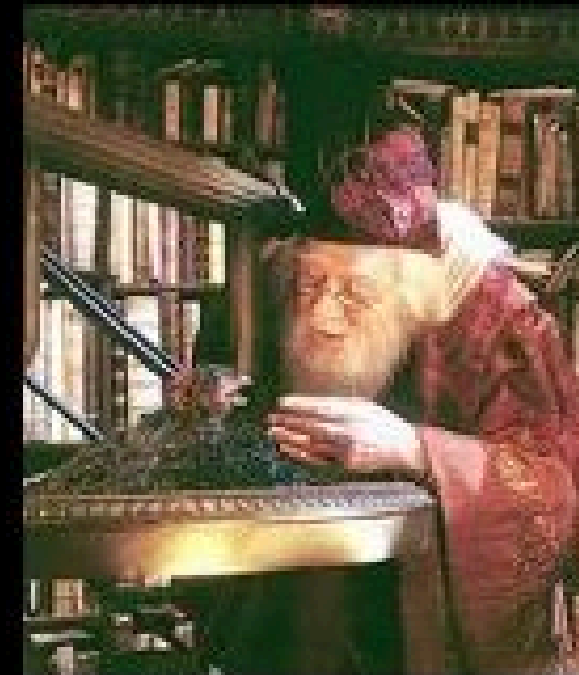
Extract topics from text using [LDA](#).

Statisticians

STATISTICIAN

$$\begin{aligned} & \int_0^{2\pi} \int_0^a \frac{r}{\sqrt{a^2-r^2}} dr d\phi \\ &= a \int_0^{2\pi} \int_0^a \frac{r}{\sqrt{a^2-r^2}} dr d\phi \\ &= a \int_0^{2\pi} \left[-\sqrt{a^2-r^2} \right]_0^a d\phi \\ &= a \int_0^{2\pi} \left[(-\sqrt{a^2-a^2}) - (-\sqrt{a^2-0}) \right] d\phi \\ &= a \int_0^{2\pi} \left[\sqrt{a^2} \right] d\phi = a \int_0^{2\pi} a d\phi \\ &= a^2 \int_0^{2\pi} d\phi = a^2 2\pi/5 \end{aligned}$$

What My Friends Think I Do



What My Colleagues Think I Do



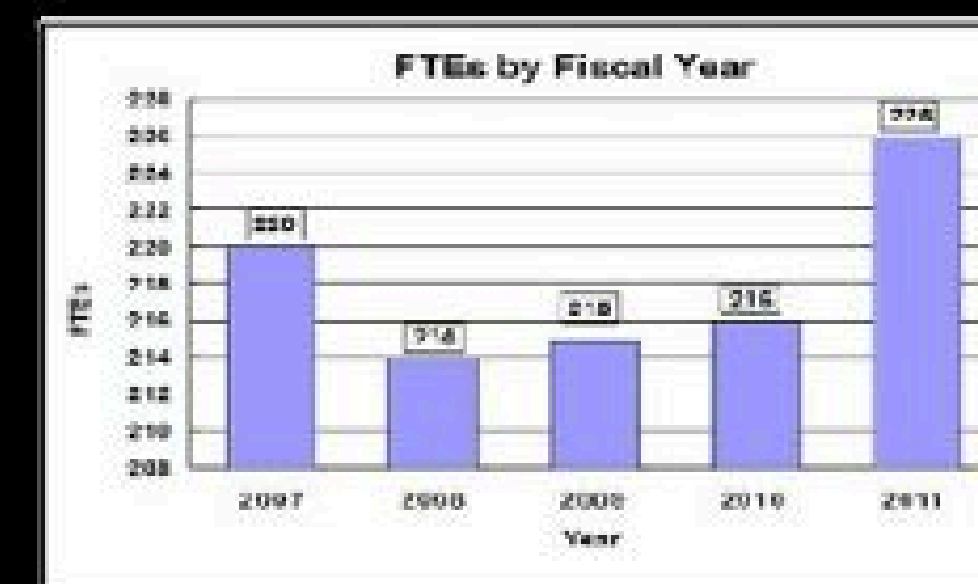
What My Parents Think I Do



What Society Thinks I Do

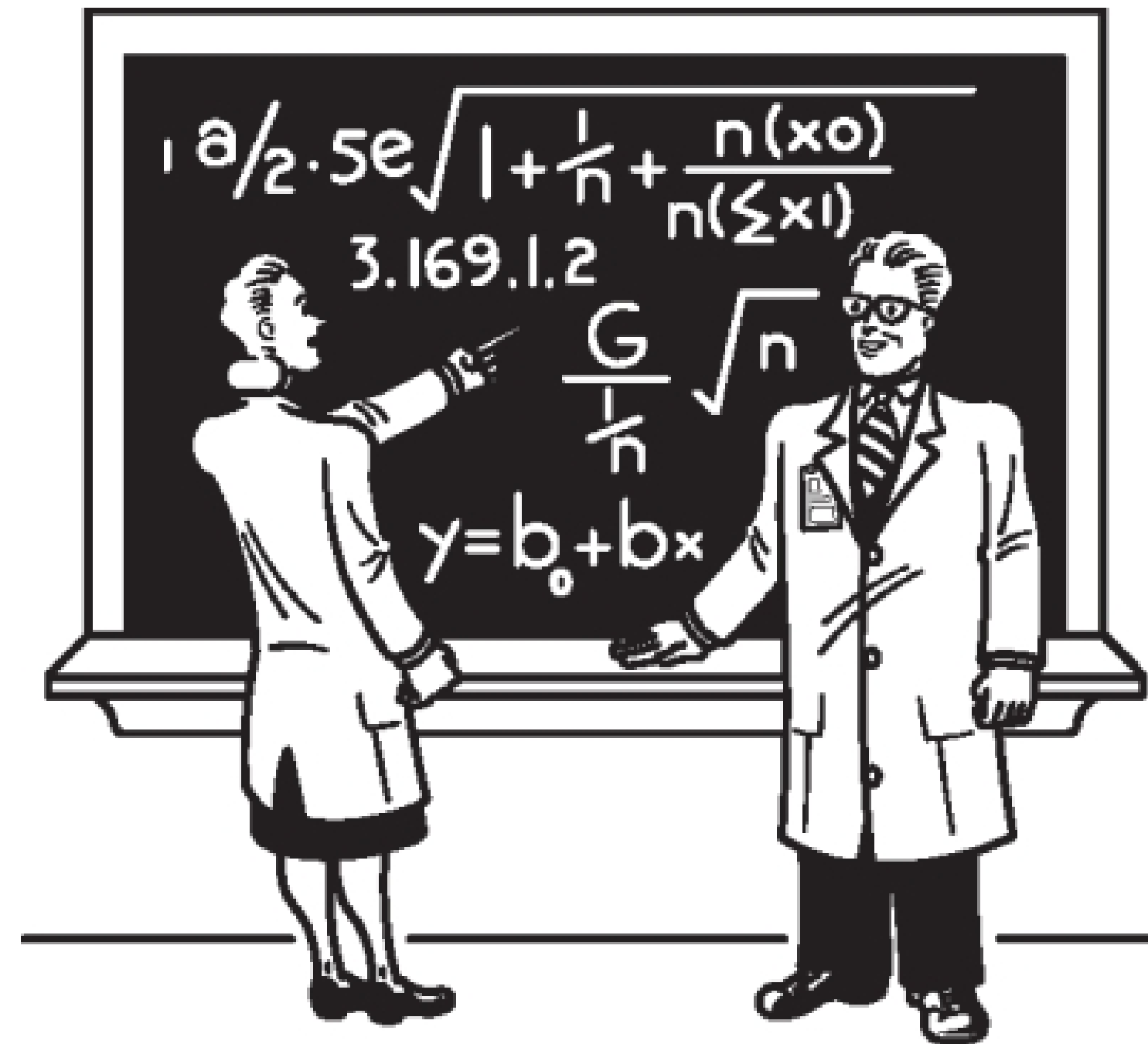


What I Think I Do



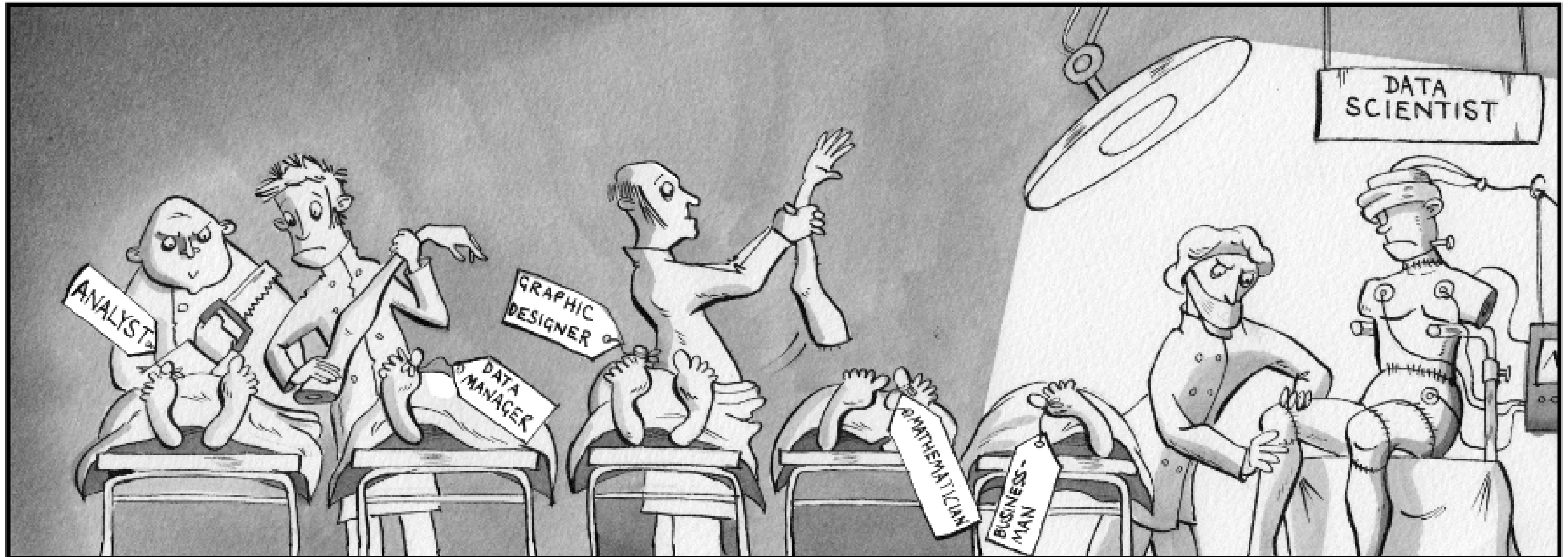
What I Actually Do

Quants



Does G stand for force due to gravity or greed?

Data scientist



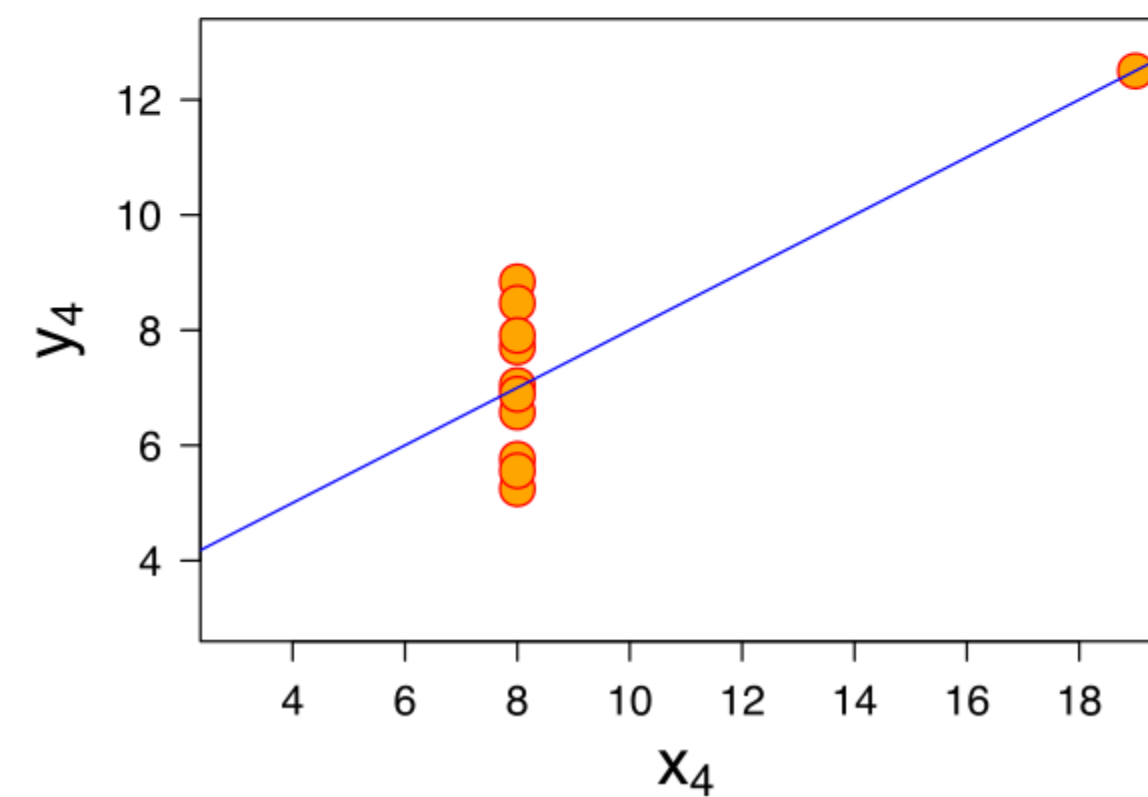
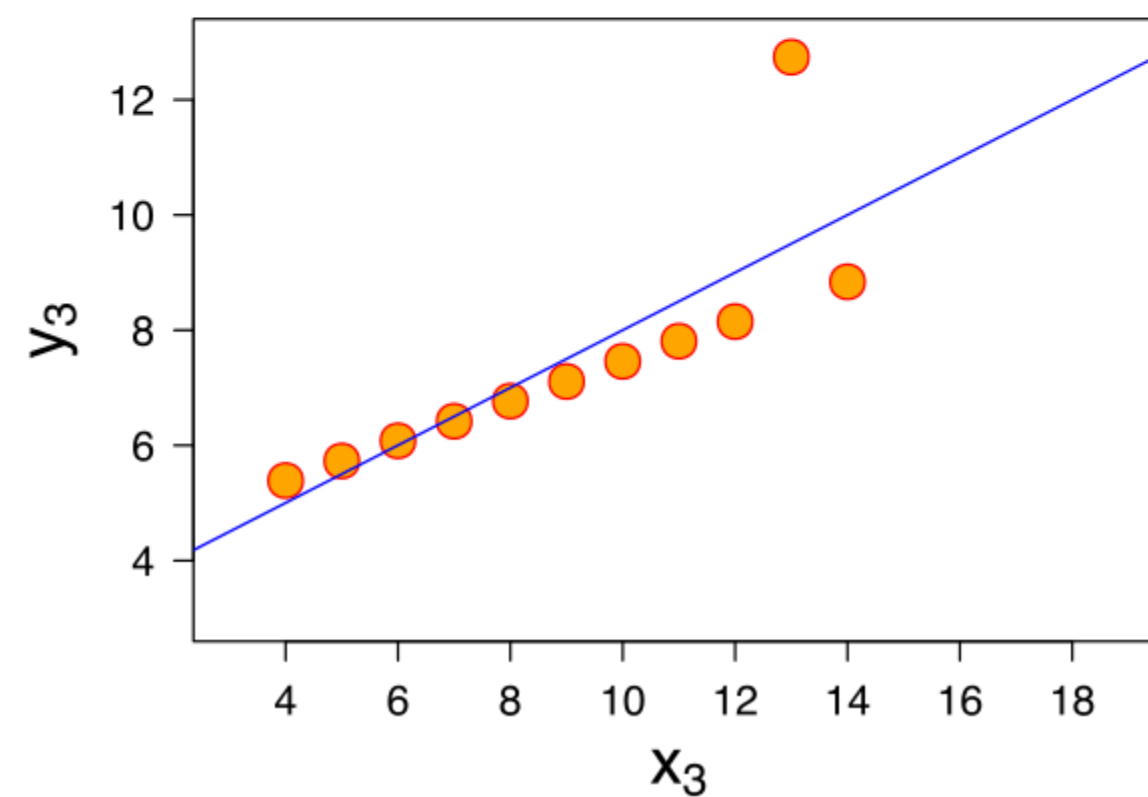
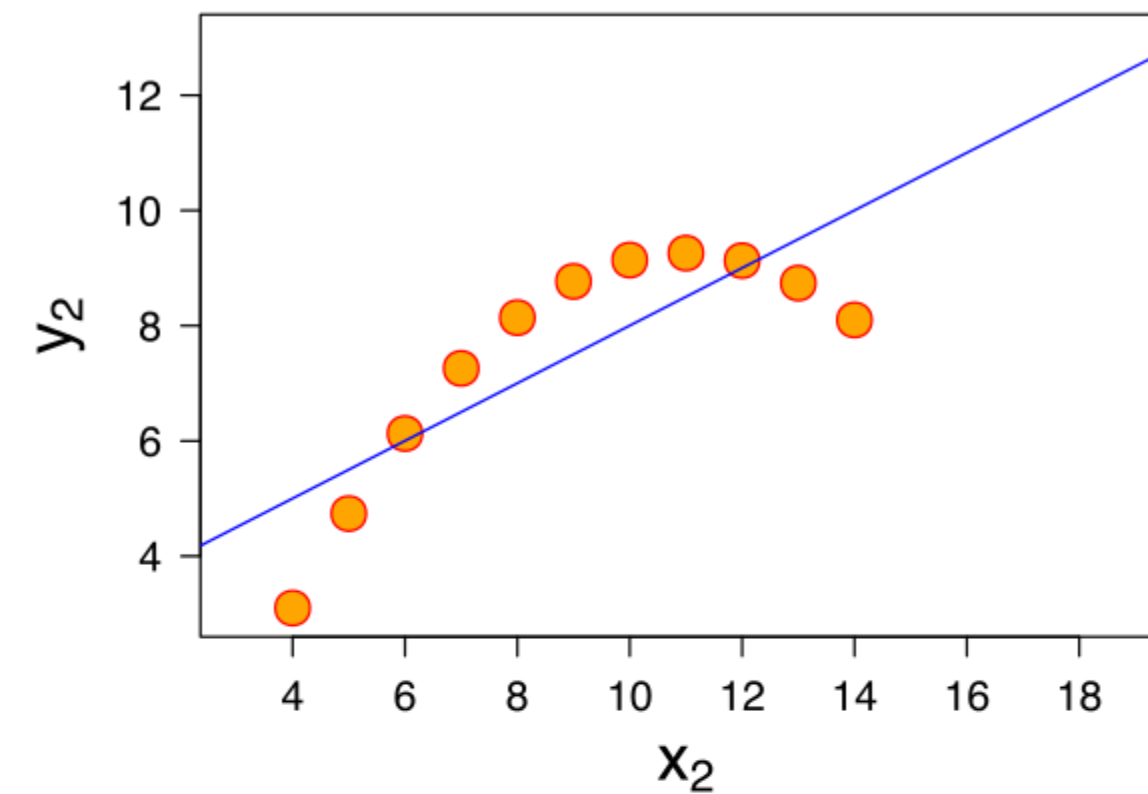
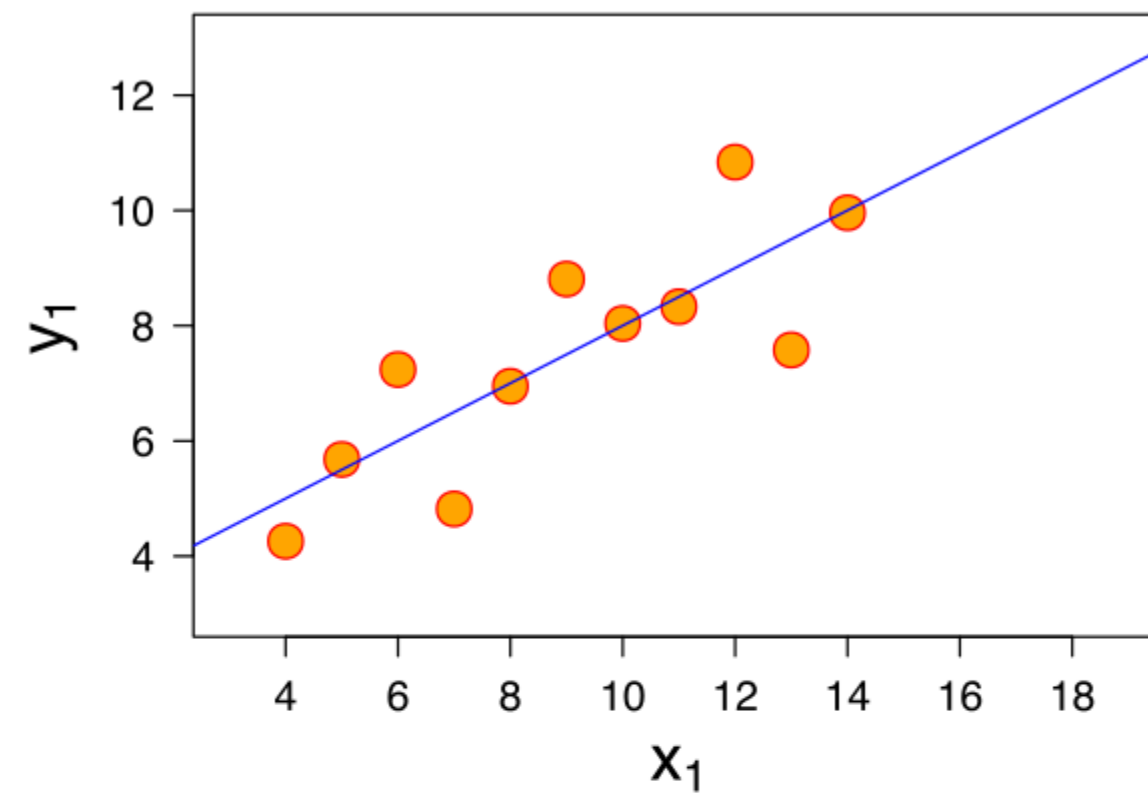
It's also important to know your data

Whatever we call our 'experts'



Who's heard of Anscombe's quartet?

Same statistical properties, but...



http://en.wikipedia.org/wiki/Anscombe's_quartet

Performance

Don't agonise over distros



*The performance of Hadoop distros
are all the same to within 1 server
within a cluster*

Stefan Groschupf
One of the creators of Hadoop

Small is still beautiful

Map Reduce without Hadoop



- Take a store (database) of 1 billion trades (a decade's worth)
- Each one is about 5k in size (so 5TB)
- Convert them to Java with binding and we hit 25k each (25TB)
- Now add some indices and we're looking at 40+TB
- Compact them with to SDOs and we now have 500GB
- Put it into Oracle Coherence V12.2.1 with lambdas and we can do

```
cache.filter(predicate).map(price).reduce(...);
```

Because latency

Latency Numbers Every Programmer Should Know

■ 1 ns

■ L1 cache reference: 0.5 ns

■ Branch mispredict: 5 ns

■ L2 cache reference: 7 ns

■ Mutex lock/unlock: 25 ns

■ = 100 ns

■ Main memory reference: 100 ns

■ = 1 μ s

■ Compress 1 KB with Zippy: 3 μ s

■ = 10 μ s

■ Send 1 KB over 1 Gbps network: 10 μ s

■ SSD random read (1Gb/s SSD): 150 μ s

■ Read 1 MB sequentially from memory: 250 μ s

■ Round trip in same datacenter: 500 μ s

■ = 1 ms

■ Read 1 MB sequentially from SSD: 1 ms

■ Disk seek: 10 ms

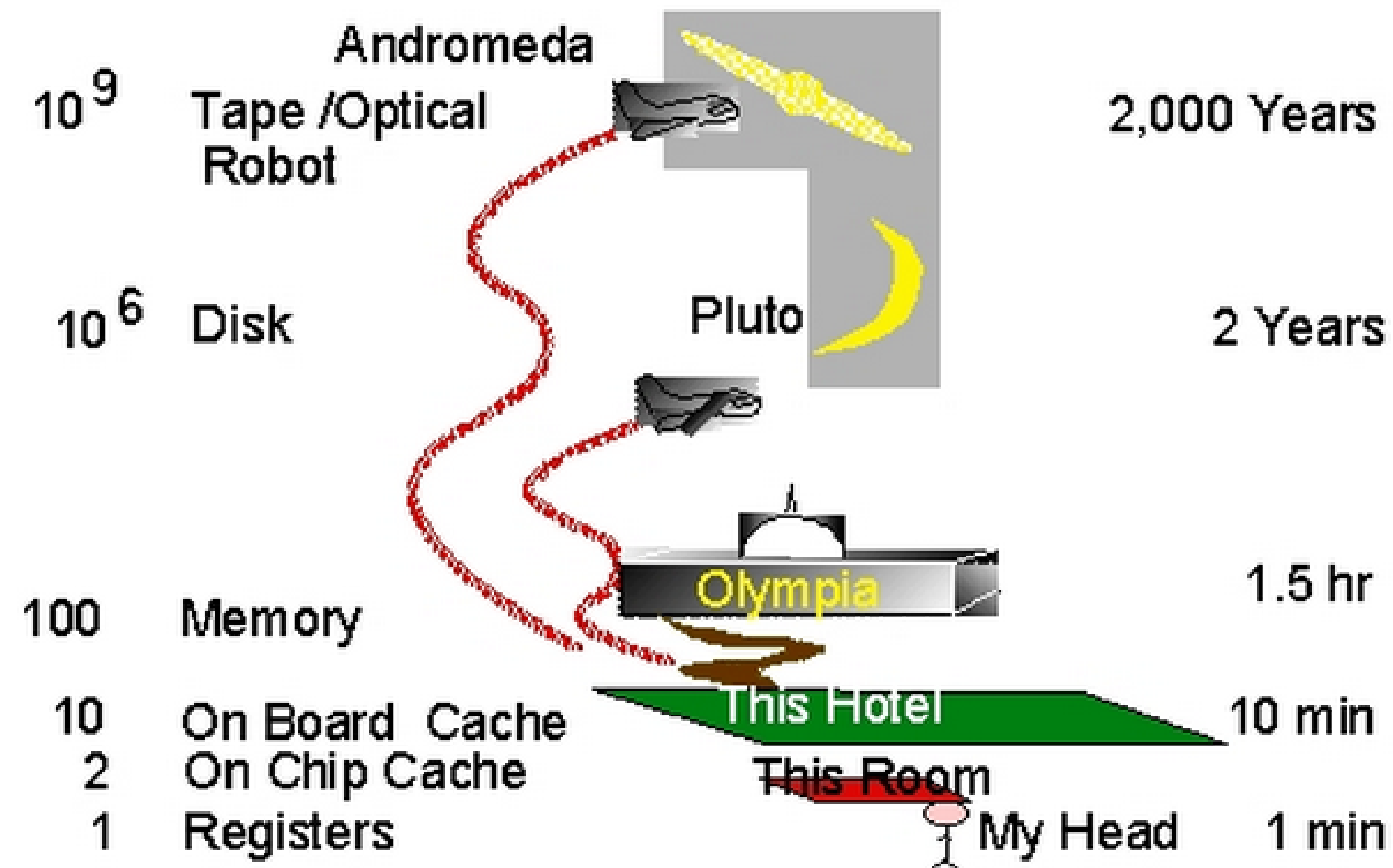
■ Read 1 MB sequentially from disk: 20 ms

■ Packet roundtrip CA to Netherlands: 150 ms

Source: <https://gist.github.com/2841832>

In terms of distance

Storage Latency: How Far Away is the Data?



23

<http://loci.cs.utk.edu/dsi/netstore99/docs/presentations/keynote/sld023.htm>

Interactive > Real time

Questions?